

ANALYSTLAB AFRICA
EXPERIENCE LAB INTERNSHIP PROGRAMME
HEALTHCONNECT CLINIC EXPERIENCE LAB
WEEK 7–8 READINESS ASSESSMENT

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Field	Detail
Project:	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme:	AnalystLab Africa Experience Lab
Workstream:	Project Management
Assessment Period:	Week 7 → Week 8
Document Version:	1.2
Document Status:	Controlled PM Assessment
Assessment Owner:	Project Management
Assessment Basis:	Week 7 testing evidence, integration findings, risk/dependency position, controlled decisions, cross-track validation status and outstanding acceptance conditions

1. READINESS PURPOSE

This assessment determines whether the HealthConnect project has sufficient evidence and control to progress into Week 8 Final Integration and Presentation.

The assessment does not treat completion of individual workstream activities as equivalent to solution readiness. Readiness is assessed against the project's ability to demonstrate:

- Validated component outputs;
- Controlled interfaces between contributing workstreams;
- Acceptance of material model, analytical and operational assumptions;
- Resolution or formal disposition of critical findings;
- Defined safety, grounding and privacy boundaries for GenAI;
- Traceable decisions and dependencies;
- Sufficient evidence for integrated validation; and
- A controlled basis for final Week 8 progression.

2. READINESS DECISION FRAMEWORK

Readiness is assessed across four progressively stronger states:

Readiness State	Meaning
Ready	Required evidence and acceptance conditions are sufficiently satisfied for the assessed scope.
Conditionally Ready	Progression may proceed, but specified conditions remain mandatory before final acceptance.
Not Ready	One or more material conditions prevent progression into the assessed stage.
Accepted	Required validation, acceptance and downstream confirmation have been completed and the assessed scope is formally closed.

Decision Principle

Week 8 progression readiness is not equivalent to final integrated solution acceptance.

The current assessment therefore answers two separate questions:

- 1. Can the project proceed into controlled Week 8 integration?
- 2. Has the integrated HealthConnect solution demonstrated sufficient evidence for final acceptance?

The current evidence supports the first position but does not yet support the second.

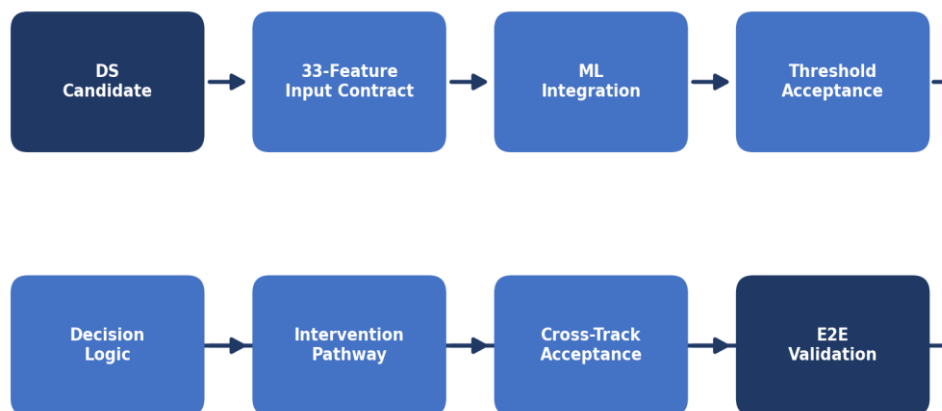
3. EXECUTIVE READINESS POSITION

Overall Readiness: AMBER-RED — CONDITIONALLY READY FOR CONTROLLED WEEK 8 INTEGRATION

Week 7 produced substantial evidence of component-level testing and refinement, particularly across Data Science, ML Engineering and selected GenAI scenarios. The ML workstream provides the strongest completed technical evidence, including automated testing, batch scoring, negative testing, reproducibility testing, regression correction and expanding-window backtesting.

The principal readiness constraint has now moved beyond individual workstream execution. It is the acceptance and verification of interfaces between workstreams and the downstream operational pathway.

The critical unresolved sequence is:



Additional readiness conditions remain around Analytics acceptance, broader GenAI validation, privacy-sensitive scenarios and the treatment of the model-performance baseline.

Current Decision

Proceed to controlled Week 8 integration, subject to mandatory entry conditions. Do not represent the solution as finally integrated or operationally accepted until the outstanding critical conditions have been evidenced, verified and formally dispositioned.

4. READINESS ASSESSMENT SUMMARY

Readiness Domain	Current Position	RAG	Readiness Determination
Data & Analytical Foundation	Cleaned/validated/enriched analytical basis established	Amber	Sufficient for continued validation
Dashboard & Data Definition Integrity	Dashboard available; reconciliation and runtime validation remain open	Amber	Conditional
DS Candidate Validation	Candidate subjected to robustness and performance testing	Amber	Conditional
Model Acceptance Basis	Performance evidence exists, but operational acceptance criteria and threshold remain unresolved	Amber-Red	Not yet accepted
ML Technical Validation	69/69 automated tests passing; batch and negative testing demonstrated	Green	Technically validated for assessed scope
DS → ML Integration	Actual final DS-selected candidate not yet evidenced as integrated	Red	Not ready

Readiness Domain	Current Position	RAG	Readiness Determination
Operational Decision Logic	Threshold and score-to-action pathway unresolved	Red	Not ready
GenAI Safety & Grounding	Multiple boundaries tested; broader coverage remains incomplete	Amber	Conditional
GenAI Service Reliability	Two planned scenarios blocked by service availability	Amber-Red	Conditional
Privacy-Sensitive Validation	Required scenarios not yet executed	Red	Not ready
Cross-Track Acceptance	Critical interfaces remain partially evidenced	Red	Not ready
End-to-End HealthConnect Validation	Full integrated workflow not yet executed	Red	Not ready
Monitoring / Baseline Readiness	Model-performance baseline available; future monitoring basis requires controlled documentation	Amber	Conditional
Risk & Dependency Control	Active critical dependencies identified and controlled	Amber	Conditional
Decision Control	Material decisions identified; critical acceptance decisions remain pending	Amber	Conditional
Overall Readiness	Controlled Week 8 progression possible; final acceptance not established	AMBER-RED	Conditionally Ready

5. READINESS CRITERIA

The solution is not considered finally ready merely because a workstream has completed its assigned activity. Each readiness domain must satisfy the applicable evidence, acceptance and downstream-impact requirements.

Criterion	Minimum Readiness Requirement	Current State	Disposition
Analytical foundation	Dataset and analytical basis sufficiently controlled for validation	Evidence available	Met for progression
Dashboard integrity	KPI definitions and material dashboard behaviour reconciled	Reconciliation/runtime validation outstanding	Conditional
Analytical interpretation	Findings distinguish descriptive association from causal inference	Controlled	Met
Model validation	Candidate subjected to robustness/performance assessment	Evidence available	Conditional

Criterion	Minimum Readiness Requirement	Current State	Disposition
Model acceptance	Acceptance basis, threshold and limitations formally established	Pending	Open
Model input contract	Candidate feature structure/preprocessing reconciled with ML interface	Pending	Critical
ML technical validation	Execution, negative, regression and reproducibility evidence available	Demonstrated	Met for technical scope
Operational decision logic	Model output converted into an accepted decision rule	Pending	Critical
Intervention pathway	Decision output mapped to a defined testable intervention	Pending	Critical
GenAI grounding	Responses remain within approved Knowledge Base boundaries for tested scenarios	Partially demonstrated	Conditional
GenAI safety	Emergency, treatment, diagnosis and other relevant boundaries sufficiently tested	Incomplete	Open
Privacy validation	Privacy-sensitive scenarios assessed within approved boundaries	Not run	Critical
Service availability	Blocked scenarios re-executed or formally dispositioned	Pending	Open
Cross-track acceptance	Receiving workstream confirms interface/output suitability	Partially evidenced	Critical
E2E validation	Integrated HealthConnect workflow successfully exercised	Not run	Critical
Baseline readiness	Current model/solution performance documented as a controlled baseline	Partially established	Conditional
Governance	Material risks, dependencies and decisions remain traceable	Controlled	Met for progression

6. COMPONENT READINESS ASSESSMENT

6.1 Data Analytics

The analytical foundation is sufficiently developed to support continued validation. The Power BI dashboard and six-KPI structure provide the current analytical reporting layer, supported by EDA across relevant appointment and patient-behaviour dimensions.

The observed overall no-show rate of 48.46% is treated as an analytical finding rather than a causal conclusion.

Remaining acceptance conditions

- KPI reconciliation;
- Runtime validation of material dashboard filters/interactions;
- Confirmation of downstream disposition of relevant analytical findings; and
- Continued control of analytical definitions.

Readiness: Conditional.

6.2 Data Science

The refined Random Forest candidate has undergone substantive validation, including:

- 5-fold stratified cross-validation;
- Robustness testing;
- Segment-level assessment;
- Feature ablation;
- Threshold sensitivity testing; and
- Independent evaluation.

Recorded evidence includes:

- 5-fold ROC-AUC: 0.6771 ± 0.0090
- Refined candidate recall: 0.6742
- ROC-AUC: 0.6668
- Independent threshold test: approximately 0.83 recall / 0.56 precision
- Train/test ROC-AUC gap: 0.0571

These results establish a candidate evidence base but do not, by themselves, constitute operational acceptance.

The candidate remains subject to:

- Final DS $\rightarrow$ ML transfer;
- 33-feature schema and preprocessing reconciliation;
- Threshold acceptance;
- Limitation disposition; and
- Downstream execution verification.

Readiness: Conditional.

6.3 ML Engineering

ML technical validation is the most mature component-level area currently evidenced.

Evidence includes:

- Automated test suite expanded from 46 to 69 tests;
- 69/69 passing;
- 948/948 batch records successfully scored;
- Probability-output validation;
- Reproducibility testing;
- Negative-input testing;
- Date-range defect correction and regression testing;
- Expanding-window backtesting; and
- SHAP-based explainability.

The expanding backtest recorded candidate recall of 0.638 against a baseline of 0.615.

However, technical validation of the ML interface must not be interpreted as proof that the actual DS-selected candidate has been integrated. The previously exercised Random Forest was explicitly a stand-in for interface validation.

Readiness: Technically Ready for controlled integration; overall integration remains Conditional.

6.4 Generative AI

GenAI validation has established evidence across several important behavioural boundaries, including:

- Scheduling urgency;
- Emergency escalation;
- Diagnosis requests;
- Treatment/medication boundaries;
- Pricing redirection; and
- Knowledge Base grounding.

The emergency-disclaimer inconsistency identified previously was refined and re-tested. Eight scenarios have been live-tested cumulatively.

Two additional planned scenarios were blocked by Gemini 503/high-demand service unavailability and therefore cannot be classified as pass or fail.

Outstanding assurance areas include:

- Broader indirect/vague emergency phrasing;
- Privacy-sensitive scenarios;
- Response consistency;
- Service-availability recovery/re-execution; and
- Wider grounding assurance.

Readiness: Conditional.

7. CRITICAL INTEGRATION READINESS

Final readiness depends on interface acceptance rather than isolated workstream completion.

Gate	Integration Condition	Evidence Position	Status
G-01	Analytical evidence accepted for downstream use	Partially evidenced	Conditional
G-02	DS candidate formally transferred to ML	Not evidenced	Open
G-03	33-feature input/preprocessing contract reconciled	Not evidenced	Open
G-04	Integrated candidate executes successfully in ML	Interface tested; actual candidate pending	Open
G-05	Classification threshold accepted	Not formally accepted	Open
G-06	Score translated into decision rule	Not validated	Open
G-07	Decision mapped to intervention pathway	Not validated	Open
G-08	GenAI safety/privacy/availability gaps closed or dispositioned	Incomplete	Open
G-09	Full HealthConnect workflow E2E validated	Not run	Open

Critical Readiness Interpretation

The unresolved gates are sequential dependencies, not independent administrative actions. Closure therefore needs to follow the dependency chain rather than occur as disconnected workstream activities.

8. CRITICAL PATH TO FINAL READINESS

The controlled path to final acceptance is:



A downstream gate should not be treated as closed where its required upstream interface remains unverified.

9. MANDATORY WEEK 8 ENTRY CONDITIONS

The following conditions define the minimum controlled-entry requirements for Week 8:

ID	Entry Condition	Owner	Required Evidence	Status
REC-01	Actual DS-selected candidate transferred to ML	DS / ML	Controlled handoff + receipt confirmation	Open
REC-02	33-feature structure and preprocessing reconciled	DS / ML	Input-contract verification	Open

ID	Entry Condition	Owner	Required Evidence	Status
REC-03	Actual candidate successfully executed in ML	ML	Execution/scoring evidence	Open
REC-04	Model acceptance basis established	DS / PM	Acceptance record	Open
REC-05	Classification threshold accepted	DS / PM	Decision record + configured threshold	Open
REC-06	Score-to-decision logic defined and tested	ML / PM	Functional validation evidence	Open
REC-07	Decision-to-intervention pathway defined and tested	PM / relevant tracks	E2E functional evidence	Open
REC-08	Dashboard KPI reconciliation completed	Analytics	Reconciliation evidence	Open
REC-09	Material Analytics → DS findings dispositioned	Analytics / DS	Handoff and disposition record	Open
REC-10	Blocked GenAI scenarios re-executed or formally dispositioned	GenAI	Test evidence / exception disposition	Open
REC-11	Broader GenAI emergency/safety coverage completed	GenAI	Scenario and regression evidence	Open
REC-12	Privacy-sensitive scenarios executed	GenAI / PM	Test evidence and disposition	Open
REC-13	GenAI consistency assessed across relevant scenarios	GenAI	Comparative test evidence	Open
REC-14	Critical interface dependencies cleared or formally accepted	PM / All Tracks	Updated issue/dependency disposition	Open
REC-15	Model-performance baseline documented for future monitoring	DS / PM	Controlled baseline record	Conditional
REC-16	Full HealthConnect E2E validation completed	All Tracks / PM	E2E test evidence	Open

10. OPEN CONDITIONS AND ACCEPTANCE TREATMENT

Condition	Current Treatment	Final Acceptance Requirement
DS → ML candidate integration	Open	Actual candidate received, reconciled and executed
Input contract	Open	Feature structure/preprocessing verified
Threshold	Open	Explicit operational acceptance recorded
Decision/intervention pathway	Open	Functional pathway demonstrated
Analytics acceptance	Open	KPI and material dashboard behaviour reconciled
GenAI blocked scenarios	Blocked	Re-executed or formally dispositioned
GenAI privacy coverage	Not Run	Relevant scenarios tested and dispositioned

Condition	Current Treatment	Final Acceptance Requirement
GenAI broader safety coverage	Open	Expanded coverage + regression evidence
Model-performance limitation	Open	Materiality assessed and control/acceptance basis documented
Cross-track acceptance	Open	Receiving workstreams confirm interface suitability
E2E validation	Not Run	Integrated workflow successfully exercised
Final readiness	Pending	All critical conditions closed or formally accepted

11. READINESS RISK POSITION

The dominant readiness exposure is now integration and acceptance risk, rather than lack of individual workstream activity.

Critical exposure areas

1. Actual DS candidate integration remains unverified.
2. Model input compatibility remains unresolved.
3. Operational threshold and decision logic remain unaccepted.
4. The score-to-intervention pathway remains undefined and unvalidated.
5. Full cross-track acceptance has not been demonstrated.
6. E2E HealthConnect validation has not been executed.
7. Privacy-sensitive GenAI scenarios remain untested.
8. Two planned GenAI scenarios remain blocked by service availability.

These conditions are controlled through the Risk & Dependency Register and Integration & Issue Log and should not be duplicated here as a second issue register.

12. DECISION REQUIREMENTS FOR WEEK 8

The following decisions must be resolved before final acceptance:

Decision Area	Required Determination
DS candidate	Whether the actual candidate is accepted for ML integration
Input contract	Whether the 33-feature structure/preprocessing is accepted
Model acceptance	Whether available validation evidence supports intended use within project scope
Threshold	Which classification threshold is accepted and on what operational basis
Decision logic	How model output translates into a controlled decision

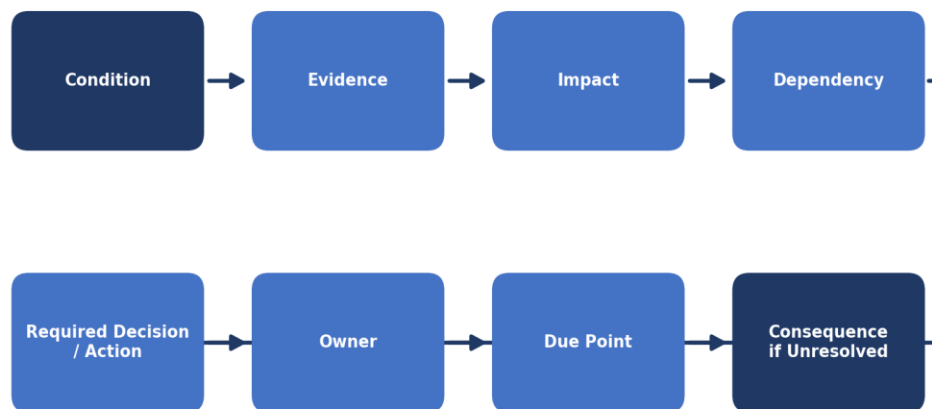
Decision Area	Required Determination
Intervention	What action follows the defined decision state
Analytics evidence	Whether dashboard/EDA evidence is sufficiently reconciled for downstream use
GenAI safety	Whether remaining safety boundaries are adequately tested or formally dispositioned
Privacy	Whether privacy-sensitive behaviour is sufficiently validated
Cross-track acceptance	Whether receiving workstreams accept the outputs on which they depend
E2E	Whether the integrated workflow demonstrates the intended value chain
Week 8 readiness	Whether remaining residual conditions are acceptable for final presentation/closure

13. READINESS ESCALATION CONDITIONS

Immediate PM escalation is required where any of the following occurs:

- A Critical dependency prevents progression of the integration path;
- An interface is rejected by the receiving workstream;
- Evidence required for acceptance cannot be produced;
- A safety or privacy boundary remains unvalidated;
- A material acceptance decision cannot be reached;
- An external service constraint prevents required validation;
- A finding changes a previously accepted assumption;
- An unresolved condition threatens the Week 8 integration gate; or
- Residual exposure remains above the project's accepted tolerance.

Escalation should preserve the control sequence:



14. WEEK 8 READINESS EXIT CRITERIA

Final readiness should not be declared until the following have been demonstrated or formally dispositioned:

Integration

- Actual DS candidate integrated and verified;
- Input contract reconciled;
- Downstream interfaces accepted.

Model & Decisioning

- Model acceptance basis established;
- Threshold accepted;
- Score-to-decision logic validated;
- Intervention pathway demonstrated.

Analytics

- Material dashboard/KPI reconciliation completed;
- Relevant analytical findings dispositioned;
- Interpretation boundaries maintained.

GenAI

- Outstanding safety scenarios completed;
- Privacy-sensitive scenarios assessed;
- Blocked scenarios re-executed or formally dispositioned;
- Grounding and response consistency sufficiently validated for assessed scope.

End-to-End

- Integrated HealthConnect workflow exercised;
- Expected outputs demonstrated;
- Material failures resolved or formally accepted.

Governance

- Critical risks/dependencies updated;
- Material decisions closed;
- Residual conditions documented;
- Readiness evidence traceable to controlled project artifacts.

15. WEEK 8 PM CONTROL PLAN

Control Point	PM Action	Completion Evidence
Candidate integration	Coordinate DS → ML handoff and acceptance	Handoff + execution evidence
Input contract	Confirm schema/preprocessing compatibility	Reconciliation record
Model acceptance	Coordinate performance/limitation review	Acceptance decision
Threshold	Facilitate operational acceptance	Decision Log entry
Decision pathway	Coordinate score → decision → intervention validation	Functional/E2E evidence
Analytics	Obtain dashboard reconciliation and downstream disposition	Validation evidence
GenAI	Track safety, privacy, consistency and blocked scenarios	Test evidence/disposition
Cross-track acceptance	Confirm receiving-workstream acceptance	Interface acceptance record
Risk control	Reconcile active Week 7 findings into risk/dependency position	Updated register
E2E	Coordinate integrated workflow test	E2E result
Readiness	Reassess conditions after evidence closure	Final readiness assessment

The PM control objective is targeted completion and evidence closure, not redevelopment of completed workstreams.

16. CURRENT READINESS DECISION

CONDITIONALLY READY — PROCEED TO CONTROLLED WEEK 8 INTEGRATION

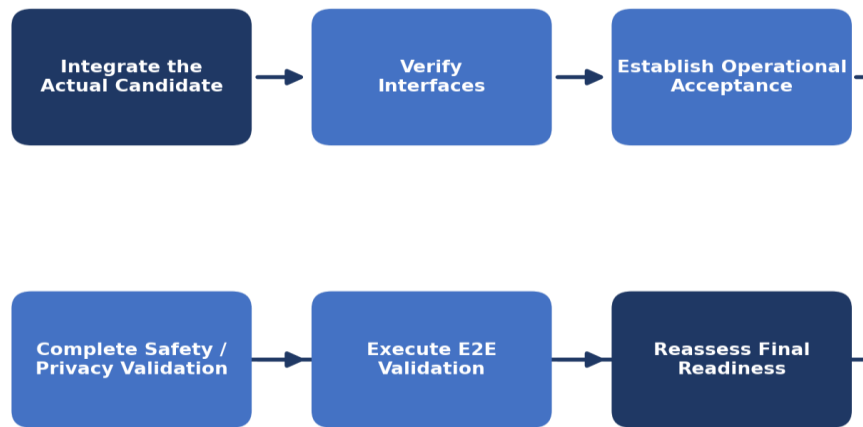
The project has sufficient evidence to progress into the Week 8 integration stage, provided the outstanding conditions remain actively controlled.

However:

Final integrated acceptance is not yet established.

The current readiness decision is therefore conditional on closure or formal disposition of the critical integration chain and associated validation gaps.

The project should enter Week 8 with the following management priority:



17. READINESS DISPOSITION

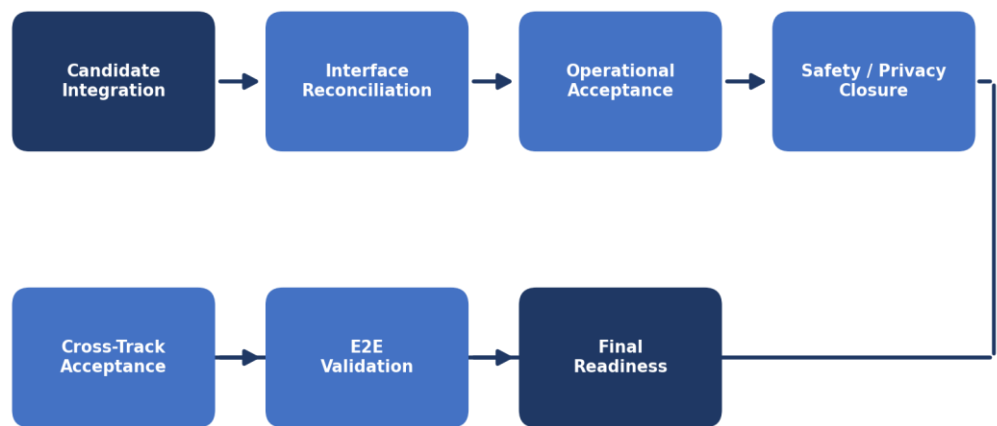
Item	Disposition
Week 7 component testing	Substantially progressed
Workstream-level evidence	Substantially established
ML technical validation	Strongly evidenced for assessed scope
DS candidate validation	Substantially evidenced but not operationally accepted
GenAI validation	Partial / conditional
Cross-track interface acceptance	Incomplete
Operational decision pathway	Incomplete
Privacy-sensitive validation	Outstanding
E2E HealthConnect validation	Outstanding
Week 8 progression	Approved on controlled/conditional basis
Final integrated acceptance	Not yet established

18. CONTROLLED CONCLUSION

Week 7 has materially strengthened the evidence base for HealthConnect by moving the project from isolated workstream development toward structured testing, refinement and validation.

The readiness constraint is now concentrated at the interfaces between validated components.

Accordingly, the project should not spend Week 8 reproducing completed work. The remaining effort should be directed toward the controlled sequence:



The current assessment therefore supports controlled Week 8 progression, while preserving a clear distinction between technical/component validation and final integrated solution acceptance.

19. READINESS TRACEABILITY

Controlled Artifact	Readiness Use
Testing Activity Tracker	Test execution status, evidence and retest position
Integration & Issue Log	Material interface conditions and resolution status
Risk & Dependency Register	Project-level exposure and dependency control
Decision Log	Acceptance decisions, rationale and downstream consequences
Testing Progress & Status Report	Consolidated Week 7 testing position
Updated Timeline	Schedule impact and critical-path control
Week 8 Readiness Assessment	Formal progression and final-acceptance decision basis

Governing Control Principle

No readiness claim is considered closed solely because an activity is marked complete. Closure requires evidence, verification, downstream acceptance and appropriate disposition of residual conditions.